

1. What is C++?

C++ is a programming language used to create different types of software, applications, games, operating systems, and many other programs.

A C++ program is written as **source code**. The computer then converts this code into a form that the computer can understand and execute.

2. My First C++ Program

Let us start with a very simple program:

```
#include <iostream>

using namespace std;

int main()
{
    cout << "Hello, World!";

    return 0;
}
```

Output

```
Hello, World!
```

This is usually the first program, write in C++.

3. Understanding the Program

Let's understand **each line** step by step.

Line 1: #include <iostream>

```
#include <iostream>
```

What does it mean?

#include tells C++ to **include a library** in our program.

iostream stands for:

Input/Output Stream

It provides facilities for taking input from the user and displaying output on the screen.

For example, we need `iostream` to use:

```
cout
```

and

```
cin
```

Example

```
#include <iostream>
```

Without including `iostream`, we cannot normally use `cout` and `cin`.

Simple idea

Think of a **library as a toolbox**.

```
iostream = Toolbox  
cout     = Tool for displaying output  
cin      = Tool for taking input
```

4. What is `#include`?

```
#include
```

`#include` is called a **preprocessor directive**.

It tells the preprocessor to include the required library/header file before the actual compilation of the program.

Important

A preprocessor directive starts with:

```
#
```

Examples:

```
#include <iostream>
```

```
#define PI 3.14159
```

For beginners, remember:

#include is used to include a library/header file in a C++ program.

5. Line 2: `using namespace std;`

```
using namespace std;
```

Let's break it down.

`std`

`std` means **standard**.

C++ provides many standard features inside the `std` namespace.

For example:

```
std::cout
```

is the standard output object.

Instead of writing:

```
std::cout << "Hello";
```

we can write:

```
cout << "Hello";
```

when we have:

```
using namespace std;
```

Explanation

Think of `std` as a **folder** containing standard C++ tools.

```
std
├── cout
├── cin
├── string
└── endl
```

`using namespace std;` allows us to use these standard tools without repeatedly writing `std::`.

6. Line 3: `int main()`

```
int main()
```

This is one of the **most important parts** of a C++ program.

`main()` is the **starting point** of a C++ program.

When we run the program, execution starts from:

```
main()
```

What does `int` mean?

`int` means **integer**.

Here, `int` tells us that the `main()` function will return an integer value.

So:

```
int main()
```

means:

The `main` function returns an integer value.

7. What is `main()`?

```
main()
```

`main()` is a **function**.

A function is a block of code designed to perform a particular task.

The `main()` function is special because it is where the execution of a C++ program begins.

Simple example

Think of a program as a book.

```
C++ Program
  ↓
  main()
  ↓
Program execution starts here
```

8. What are `{ }`?

Our program contains:

```
{
    cout << "Hello, World!";

    return 0;
}
```

The symbols:

```
{
}
```

are called **curly braces** or **braces**.

They mark the beginning and end of a block of code.

`{`

Opening brace — starts the block.

`}`

Closing brace — ends the block.

For example:

```
int main()
{
    // Program code
}
```

Everything between { and } belongs to the `main()` function.

9. `cout << "Hello, World!";`

```
cout << "Hello, World!";
```

This statement displays text on the screen.

`cout`

`cout` is used to **display output**.

Think:

```
cout = output
```

`<<`

The symbol:

```
<<
```

is called the **insertion operator** when used with `cout`.

It sends the data toward `cout` for display.

`"Hello, World!"`

This is the text that we want to display.

Complete statement

```
cout << "Hello, World!";
```

means:

Display `Hello, World!` on the screen.

10. What are " "?

In our program:

```
"Hello, World!"
```

the text is enclosed in double quotation marks.

```
" "
```

Double quotation marks are used to represent a **string literal**.

For example:

```
cout << "Welcome";
```

Output:

```
Welcome
```

Another example:

```
cout << "My name is Ali";
```

Output:

```
My name is Ali
```

11. Why is there a semicolon ;?

Notice:

```
cout << "Hello, World!";
```

There is a semicolon at the end.

```
;
```

A semicolon is generally used to indicate the **end of a statement** in C++.

For example:

```
int age = 20;
```

```
cout << age;
```

```
return 0;
```

Important

A common mistake is forgetting the semicolon (;) at the end of a statement.

Incorrect:

```
cout << "Hello"
```

Correct:

```
cout << "Hello";
```

12. return 0;

The last statement is:

```
return 0;
```

It returns the value 0 from the `main()` function.

Remember:

```
int main()
```

means `main()` returns an integer.

Therefore:

```
return 0;
```

returns an integer value of 0.

you can understand it as:

The program has finished successfully.

13. Complete Program Structure

A basic C++ program can look like this:

```
#include <iostream>

using namespace std;

int main()
{
    // Program statements

    return 0;
}
```

Let's visualize it:

```
#include <iostream>
      ↓
Include required library

using namespace std;
      ↓
Use standard C++ features

int main()
      ↓
Program starts here

{
      ↓
Beginning of main function

Program statements
      ↓
Instructions executed by computer

return 0;
      ↓
End main function successfully

}
      ↓
End of program block
```

14. Modify Our First Program

Original program:

```
#include <iostream>

using namespace std;

int main()
{
    cout << "Hello, World!";

    return 0;
}
```

Now change the message:

```
#include <iostream>

using namespace std;

int main()
{
    cout << "Welcome to C++ Programming!";

    return 0;
}
```

Output

```
Welcome to C++ Programming!
```

15. Display Your Name

Try this program:

```
#include <iostream>

using namespace std;

int main()
{
    cout << "My name is Muhammad.";
```

```
    return 0;  
}
```

Output

```
My name is Muhammad.
```

Students can replace **Muhammad** with their own name.

16. Display Multiple Lines

We can use **endl** to move to the next line.

```
#include <iostream>  
  
using namespace std;  
  
int main()  
{  
    cout << "My name is Ali." << endl;  
    cout << "I am a student." << endl;  
    cout << "I am learning C++."  
  
    return 0;  
}
```

Output

```
My name is Ali.  
I am a student.  
I am learning C++.
```

Understanding **endl**

```
endl
```

means **end line**.

It moves the cursor to the next line.

For example:

```
cout << "Hello" << endl;
```

```
cout << "Welcome";
```

Output:

```
Hello  
Welcome
```

17. Another Way to Print Multiple Lines

We can also use `\n`.

```
#include <iostream>  
  
using namespace std;  
  
int main()  
{  
    cout << "My name is Ali.\n";  
    cout << "I am a student.\n";  
    cout << "I am learning C++.";  
  
    return 0;  
}
```

Output:

```
My name is Ali.  
I am a student.  
I am learning C++.
```

`\n` means **new line**.

18. Important C++ Terms

Term	Meaning
C++	A programming language
Program	A set of instructions
<code>#include</code>	Includes a header/library
<code>iostream</code>	Input/output library
<code>main()</code>	Starting point of the program

Term	Meaning
<code>int</code>	Integer data type
<code>cout</code>	Displays output
<code><<</code>	Sends data to <code>cout</code>
<code>{ }</code>	Defines a block of code
<code>;</code>	Ends a statement
<code>return 0</code>	Returns 0 from <code>main()</code>
<code>endl</code>	Moves output to next line
<code>\n</code>	New-line character

19. Task

Task

Write a C++ program that displays:

```
Welcome to C++ Programming
My name is _____
I am a student
I am learning programming
```

Solution

```
#include <iostream>

using namespace std;

int main()
{
    cout << "Welcome to C++ Programming" << endl;
    cout << "My name is Muhammad" << endl;
    cout << "I am a student" << endl;
    cout << "I am learning programming";

    return 0;
}
```

replace `Muhammad` with their own name.

20. Remember These 5 Things

For your **first C++ program**, remember:

1. `#include <iostream>` → allows us to use input/output features.
2. `using namespace std;` → allows us to use standard features such as `cout` without writing `std::`.
3. `int main()` → execution starts here.
4. `cout` → displays output.
5. `;` → marks the end of a statement.

First Program to Memorize

```
#include <iostream>

using namespace std;

int main()
{
    cout << "Hello, World!";

    return 0;
}
```

Best approach: first understand what each line does, then modify the message, and finally write the program yourself without looking at the example.